

WEB DASHBOARD CS 400 • Advanced Data Structures

DataStruct Explorer

Select an algorithm workspace below to launch interactive step-by-step visualizations, manipulate 2D/3D trees, and analyze execution complexities in real-time.

Structure Workspaces

5 Visualizers Active

SPATIAL PARTITIONING

Quadtree

Recursively partition 2D space into four quadrants to efficiently query spatial data.

Ins: $O(\log n)$ Src: $O(\log n)$ Del: $O(\log n)$

[Open Workspace](#)

3D SPATIAL PARTITIONING

Octree

Extend spatial partitioning to 3D — divide a cube into eight octants recursively.

Ins: $O(\log n)$ Src: $O(\log n)$ Del: $O(\log n)$

[Open Workspace](#)

PREFIX STRUCTURE

Trie (Prefix Tree)

Store strings character-by-character, sharing common prefixes for fast retrieval.

Ins: $O(L)$ Src: $O(L)$ Del: $O(L)$

[Open Workspace](#)

SELF-BALANCING BST

Splay Tree

SELF-BALANCING BST

Red-Black Tree

Compare Matrix

3D SPATIAL PARTITIONING

Octree

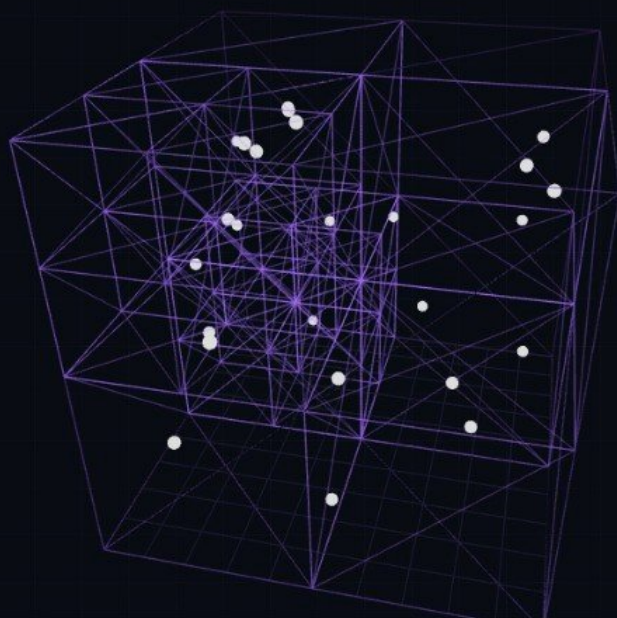
Extend spatial partitioning to 3D — divide a cube into eight octants recursively.

+ Add 3D Point ✨ Scatter 10 ⚡ Scatter 25 🗑 Clear

Insert $O(\log n)$ Search $O(\log n)$ Delete $O(\log n)$ Space $O(n)$

Capacity: 4

Drag to rotate 3D view



LIVE INSPECTOR STATS

Total 3D Points

25

Tree Depth

2

Capacity

4

Sub-Octants

25

VISUAL LEGEND

- 3D Sphere Point
- Sub-Octant Bounding Wireframe
- Root Bounding Cube

CLICK POINT TO DELETE (25)

(-2.5, -4.1, -4.1)

(-4.3, -0.4, -3.0)

(-1.7, -0.3, -3.6)

(-0.0, -0.4, -4.4)

(-4.3, -0.2, -0.1)

(4.0, -2.9, -2.1)

(3.6, -0.1, -4.4)

(1.1, -2.4, -2.0)

Use controls to step through operation animation

Power & Battery

Energy saver is on

DS

DataStruct Explorer

INTERACTIVE WEB DASHBOARD

Dashboard

Quadtrees

Otrees

Trie

Splay

RB Tree

Compare Matrix

NORMAL

SPEED

PREFIX STRUCTURE

Trie (Prefix Tree)

Store strings character-by-character, sharing common prefixes for fast retrieval.

s

Insert Word

Prefix Search: Type prefix to highlight path

Insert

0(L)

Search

0(L)

Delete

0(L)

Space

0(n × L)

Load Examples

Clear

LIVE INSPECTOR STATS

Stored Words

4

Total Nodes

12

Max Depth

3

Root Edges

4

VISUAL LEGEND

Prefix Character Node

Word Endpoint Marker

Active Search Path

STORED WORDS (4)

cat

mat

fat

at

ROOT

c

m

f

a

a

a

a

t

t

t

t